Informatica® Cloud Data Integration

Microsoft Fabric OneLake Connector

Informatica Cloud Data Integration Microsoft Fabric OneLake Connector
April 2026

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Preface

Use Microsoft Fabric OneLake Connector to learn how to read or write to Microsoft Fabric OneLake by using Data Integration. Learn to create a Microsoft Fabric OneLake connection, develop, and run mappings and mappings in advanced mode, and tasks in Data Integration.

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IDMC web site

You can access the IDMC web site at http://www.informatica.com/cloud. This site contains information about Informatica Cloud integration services.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

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The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introduction to Microsoft Fabric OneLake Connector

You can use Microsoft Fabric OneLake Connector to securely read data from or write to Microsoft Fabric OneLake.

Use Microsoft Fabric OneLake Connector to read and write flat files and flat structured complex files such as Avro, JSON, ORC, and Parquet.

You can use the Delta file format to read a delta lake or create a delta lake. You can use the Document file format in mappings in advanced mode to read PDF files.

You can use the Icerberg file format when you run a mapping to read from and write to Apache Icerberg files. You can also use the Iceberg file format when you run a mapping in advanced mode to read from Apache Iceberg files.

You can use Microsoft Fabric OneLake objects as sources and targets in mappings and mapping tasks.

You can switch mappings to advanced mode to include transformations and functions that enable advanced functionality. The advanced cluster can be hosted on the Microsoft Azure or self-service clusters.

Microsoft Fabric OneLake Connector assets

Create assets in Data Integration to integrate data using Microsoft Fabric OneLake Connector.

When you use Microsoft Fabric OneLake Connector, you can include the following Data Integration assets:

- Mapping
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

CHAPTER 2

Connections for Microsoft Fabric OneLake

Create a Microsoft Fabric OneLake connection to securely read data from or write data to Microsoft Fabric OneLake.

Before you create a connection, enable the connector for the runtime environment. For more information about enabling connectors, see *Service and connector assignment* in the Administrator help.

You can use a Microsoft Fabric OneLake connection to specify sources and targets in mappings and mapping tasks.

Prerequisites

Before you configure the connection properties, make sure to complete the following prerequisites:

1. Power BI prerequisites:
   a. Create a workspace, if it does not already exist.
   b. Create a lakehouse in the workspace. Make sure you have the required admin privileges to create a lakehouse.

   For more information about creating a workspace and lakehouse, see Creating a workspace and lakehouse.
2. Microsoft Azure portal prerequisites:
   a. Register your application in Microsoft Entra ID and generate client ID, tenant ID, and client secret. The connector does not require any API permissions assigned to your application. Additionally, you can remove any API permissions that are assigned to the application.
   b. Create a security group with assigned owners or members in Microsoft Entra ID.
   For more information about creating a security group, see Creating a security group.
3. To allow Microsoft Entra app to use the Power BI APIs, enable the Power BI service admin settings, and then add the Microsoft Entra app to the workspace in Power BI.

   Note: The Manage access pane in the Microsoft Entra app allows you to add roles based on the required capabilities. The pane displays admin, member, contributor, and viewer roles. The maximum number of capabilities are associated with the admin role. However, to access Microsoft Fabric OneLake and perform read and write operations, you need at least the Contributor role.

   For more information about enabling access in the Power BI admin portal and adding the Microsoft Entra app to the workspace, see Enabling access in the Power BI portal.

Connect to Microsoft Fabric OneLake

Let's configure the Microsoft Fabric OneLake connection properties to connect to Microsoft Fabric OneLake.

Before you begin

Before you get started, you'll need to keep the workspace name, lakehouse path, client ID, client secret, and tenant ID details handy.

Check out "Prerequisites" on page 8 to make sure that the setup is complete and all the prerequisites are in place before you configure a connection.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select which credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run the tasks. Select a Secure Agent, Hosted Agent, or serverless runtime environment. For more information about how to configure and use the runtime environments, see *Runtime Environments* in the Administrator help.
Workspace Name	Name of the workspace in Microsoft Fabric OneLake. Workspace name cannot contain special characters or spaces.
Lakehouse Path	Path or name of the lakehouse present in the workspace. You can specify the path in one of the following ways: - *root directory (/)* to access the files in the workspace. - *lakehouse name/Files* to access the files present in the lakehouse.
Authentication Type	Authentication type to access Microsoft Fabric OneLake. Service Principal Authentication uses the client ID, client secret, and tenant ID to connect to Microsoft Fabric OneLake.

Property	Description
Client ID	The application ID or client ID of your application registered in the Azure Active Directory.
Client Secret	The client secret of your application registered in the Azure Active Directory.
Tenant ID	The ID of the Azure Active Directory instance in which you created the application.
Microsoft Fabric OneLake Endpoint	The type of Microsoft Fabric OneLake endpoint that you want to connect to. Default is **fabric.microsoft.com**.

Proxy Server Settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica Intelligent Cloud Services through the proxy server.

You can configure the Secure Agent to use the proxy server on Windows and Linux. You can use the unauthenticated or authenticated proxy server.

You can use one of the following types of proxy servers:

- Unauthenticated proxy - Requires only the host and port address for configuration.
- Authenticated proxy - Requires the host address, port address, user name, and password for configuration.

To configure proxy settings for the Secure Agent, use one of the following methods:

- Configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

  For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.
- Configure the JVM options for the DTM in the Secure Agent properties. For instructions, see the Proxy server settings Knowledge article.

Bypass the proxy server

You can bypass the proxy server settings configured for the Secure Agent.

Perform the following steps to bypass the proxy server:

1. Navigate to the following directory:
   `<Secure Agent installation directory>/apps/agentcore`
2. Specify the following command in the `proxy.ini` file:
   ```
   InfaAgent.NonProxyHost=localhost|{*}onelake.blob.fabric.microsoft.com|
   onelake.dfs.fabric.microsoft.com |127.|[\:\:1]*
   ```
3. Restart the Secure Agent.

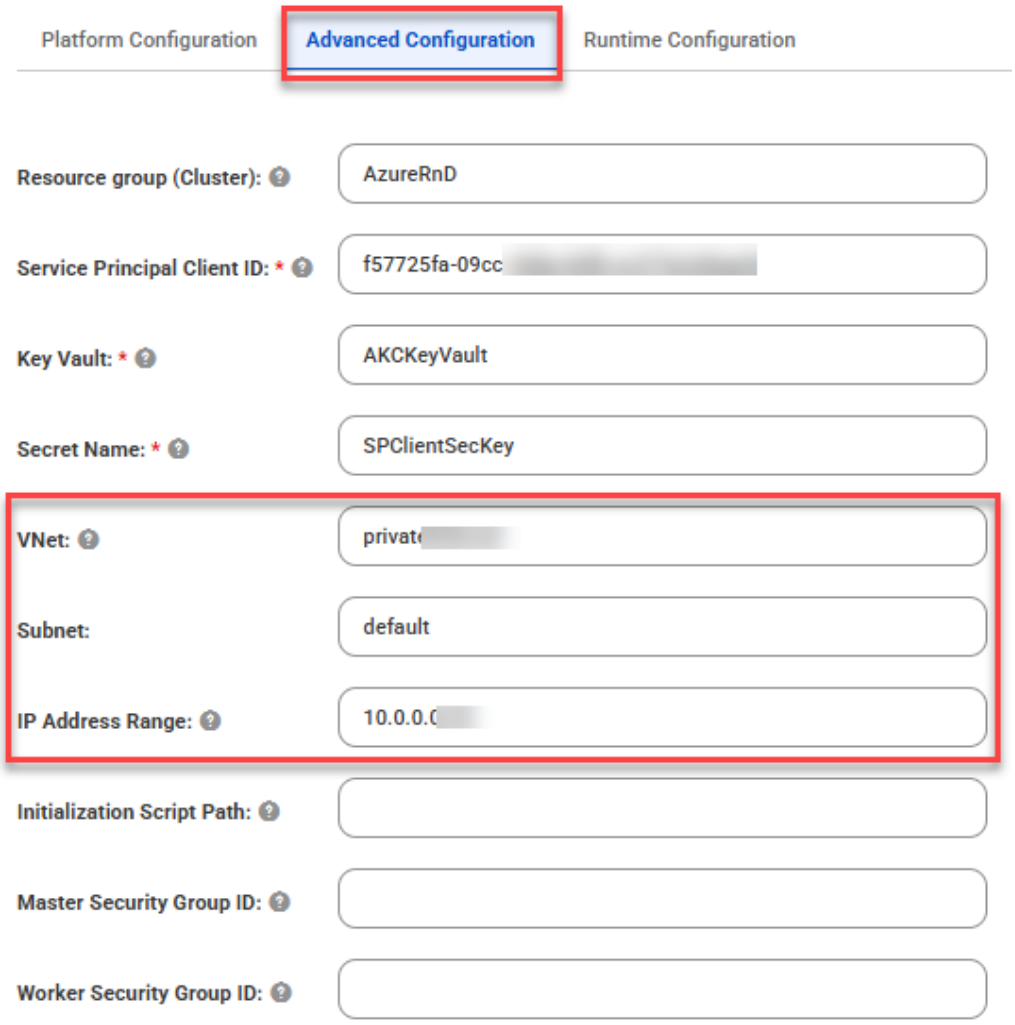
Private links to access Microsoft Fabric OneLake

You can access Microsoft Fabric OneLake using Azure Private Link endpoints.

To connect to the Microsoft Fabric OneLake account over the private Azure network, see Secure connectivity to Microsoft Fabric.

To run mappings in advanced mode over the private Azure network, additionally you must perform the following steps after you complete the required prerequisites:

1. Log in to Informatica Intelligent Cloud Services, and then click **Administrator**.
2. In **Administrator**, navigate to **Advanced Clusters** to create or modify an Advanced Configuration.
3. In the **Advanced Configuration** tab, add the Vnet, Subnet, and IP Address Range properties that is linked with the private DNS zone. This helps to spawn the cluster resources under this subnet and provide access to the private DNS zone.



CHAPTER 3

Mappings for Microsoft Fabric OneLake

When you configure a mapping, you describe the flow of data from the source to the target.

A mapping defines reusable data flow logic that you can use in mapping tasks.

When you create a mapping, you define the Source transformation and Target transformation to represent a Microsoft Fabric OneLake object. Use the Mapping Designer in Data Integration to add the Source or Target transformations in the mapping canvas and configure the Microsoft Fabric OneLake source and target properties.

In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality.

You can use Monitor to monitor the jobs.

Microsoft Fabric OneLake sources in mappings

In a mapping, you can configure a Source transformation to represent a Microsoft Fabric OneLake object.

The following table describes the Microsoft Fabric OneLake source properties that you can configure in a source transformation:

Property	Description
Connection	Name of the source connection. Select a source connection or click **New Parameter** to define a new parameter for the source connection. If you want to overwrite the parameter at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties. Ensure that the parameter file is in the correct format. When you switch between a non-parameterized and a parameterized Microsoft Fabric OneLake connection, the advanced property values are retained.
Source Type	Select Single Object or Parameter.
Object	Name of the source object. Ensure that the headers or file data does not contain special characters.

Property	Description
Parameter	Select an existing parameter for the source object or click **New Parameter** to define a new parameter for the source object. The **Parameter** property appears only if you select Parameter as the source type. When you parameterize the source object, specify the complete object path including the file system in the default value of the parameter. If you want to overwrite the parameter at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties. Ensure that the parameter file is in the correct format.
Format	The format of the file that you want to read from Microsoft Fabric OneLake. Select one of the following file formats: - Avro - JSON - ORC - Flat - Parquet - Discover Structure[1] - Iceberg - Delta. Reads data from a delta lake. To use the delta file format in a mapping in advanced mode, see "Configuring Delta file format in a mapping in advanced mode" on page 22. - Document. Reads data from PDF files.[1] Default is **None**. If you select **None** as the format type, Microsoft Fabric OneLake Connector reads data from Microsoft Fabric OneLake files in binary format. **Note:** Ensure that the source file is not empty. You cannot read a JSON file that exceeds 200 MB. For more information, see "File formatting options" on page 20.
Input Type	Determines the type of data that you can read when you select the Document file format. You can only read from PDF files with document file format.
Intelligent Structure Model[1]	Determines the underlying patterns in a sample file and autogenerates a model for files with the same data and structure. This property applies to the discover structure format type. You can either select an existing model to associate with the Source transformation, or create a new model. For more information about intelligent structure models, see *Components* in the Data Integration documentation.
[1]Applies only to mappings in advanced mode. [2]Doesn't apply to mappings in advanced mode.	

The following table describes the Microsoft Fabric OneLake source advance properties:

Property	Description
Workspace Override	Overrides the workspace specified in the connection.
Allow Wildcard Characters	Indicates whether you want to use wildcard characters for the directory source type. If you select this option, you can use the question mark (?) and asterisk (*) wildcard characters to define one or more characters in a search. For more information, see "Wildcard characters" on page 15.

Property	Description
Lakehouse Override	Overrides the lakehouse specified in the connection.
File Name Override	Overrides the source object at run time.
Source Type	The type of source from which you want to read data. Select the following source types: - File - Directory - Directory - Recursive Select **Directory - Recursive** to recursively read from flat and complex files. Default is File.
Compression Format	Reads compressed data from the source. Select one of the following options: - **None**. To read uncompressed and Parquet files. - **Gzip**. To read flat files and Parquet files that use Gzip compression. The compressed files must have the `.gz` extension. You cannot preview data for a compressed flat file.
Incremental File Load	Indicates whether you want to incrementally load files when you use a directory as the source for mappings in advanced mode. When you incrementally load files, the mapping task reads and processes only the files in the directory that have changed since the mapping task last ran. For more information, see "Incrementally loading files" on page 16.
Tracing Level	Sets the amount of detail that appears in the log file. You can choose terse, normal, verbose initialization or verbose data. Default is normal.

Directory source in Microsoft Fabric OneLake sources

You can select the type of source from which you want to read data.

You can select the following type of sources from the **Source Type** option under the advanced source properties:

- File
- Directory
- Directory - Recursive

Use the following rules and guidelines to select **Directory** as the source type:

- All the source files in the directory must contain the same metadata.
- All the files must have data in the same format. For example, delimiters, header fields, and escape characters must be same.
- All the files under a specified directory are parsed. To parse the files in the subdirectories, use recursive read.
  For more information, see "Reading files from subdirectories" on page 15.

Wildcard characters

When you read an Avro, JSON, ORC, Parquet, flat file, or PDF, you can use the `?` and `*` wildcard characters to define one or more characters in a search.

To use wildcard characters for the source file name, select the source type as **Directory** and enable the **Allow Wildcard Characters** option in the advanced source properties.

You can use the following wildcard characters:

? (Question mark)

The question mark character (?) allows one occurrence of any character. For example, if you enter the source file name as `a?b.txt`, the Secure Agent reads data from files with the following names:

- `a1b.txt`
- `a2b.txt`
- `aab.txt`
- `acb.txt`

*** (Asterisk)**

The asterisk mark character (*) allows zero or more than one occurrence of any character. If you enter the source file name as `a*b.txt`, the Secure Agent reads data from files with the following names:

- `aab.txt`
- `a1b.txt`
- `ab.txt`
- `abc11b.txt`

Rules and guidelines for wildcard characters

Consider the following rules and guidelines when you use wildcard characters:

Mappings

- You cannot use wildcard characters when you read from partition columns.
- When you read a complex file in a mapping, do not use a tilde (~) in the sub-directory name or file name.
- When you use wildcard characters in directory override, the Secure Agent reads data from the folders as well as the files that match the name pattern.

Mappings in advanced mode

- When you read a flat file or complex file and enable wildcard characters, ensure that the path specified in the directory override or file name override matches the file path in the source.
- When you use wildcard characters, ensure that the file name does not start with a special character.
- When you read a flat file, do not use the following special characters in the directory name or sub-directory name in the directory override:

  `[] {} " ' + ^ % * ? space`

Reading files from subdirectories

You can read objects stored in subdirectories in Microsoft Fabric OneLake in mappings.

You can use recursive read for flat files and complex files in mappings. When you create a mapping in advanced mode, you cannot use recursive read for flat files.

To read objects stored in subdirectories, select the source type as **Directory - Recursive** in the advanced source properties.

Rules and guidelines for reading from subdirectories

Consider the following rules and guidelines when you read objects stored in subdirectories:

Mappings

- When you read from or write to a flat file in Microsoft Fabric OneLake, ensure that the directory or subdirectory name does not contain the percentage (%) character. Else, the mapping fails.
- You cannot use recursive read when you read from partition columns.
- When you read a complex file in a mapping, do not use a tilde (~) in the subdirectory name or file name.

Mappings in advanced mode

- When you read a complex file and enable recursive read, ensure that the path specified in the directory override or file name override matches the file path in the source.
- When you read a flat file, do not use the following special characters in the directory name or sub-directory name in the directory override:

```
[] {} " '  +  ^  %  *  ?  space
```

Incrementally loading files

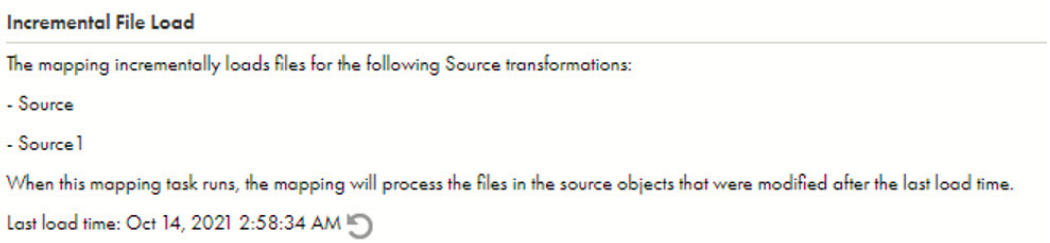
You can incrementally load source files from a directory to read and process only the files that have changed since the last time the mapping task ran.

You can incrementally load files only in mappings in advanced mode. Ensure that all of the source files exist in the same Cloud environment. You can incrementally load files when you use ORC, Flat, Parquet, JSON, Discover Structure, and Document file formats.

To incrementally load source files, select **Incremental File Load** and **Directory** or **Directory-Recursive** as the source type in the advanced read options of the Microsoft Fabric OneLake object.

When you incrementally load files from Microsoft Fabric OneLake, the job loads files that have changed from the last load time to five minutes before the job started running. For example, if you run a job at 2:00 p.m, the job loads files changed before 1:55 p.m. The five-minute buffer ensures that the job loads only complete files because uploading objects on Microsoft Fabric OneLake can take a few minutes to complete.

When you configure a mapping task, the **Incremental File Load** section lists the Source transformations that will incrementally load files and the time that the last job completed loading the files. By default, the next job that runs checks for files modified after the last load time.



You can also override the load time that the mapping uses to look for changed files in the specified source directory. You can reset the incremental file load settings to perform a full load of all the changed files in the directory, or you can configure a time that the mapping uses to look for changed files.

For more information on incremental loading, see Reprocessing incrementally-loaded source files in *Tasks* in the Data Integration documentation.

Microsoft Fabric OneLake targets in mappings

In a mapping, you can use a Microsoft Fabric OneLake object as a target.

When you use Microsoft Fabric OneLake target objects, you can select a Microsoft Fabric OneLake collection as target. You can configure Microsoft Fabric OneLake target properties on the Target page of the Mapping wizard. When you write data to Microsoft Fabric OneLake, you can use the create target field to create a target at run time. When you create a new target based on the source, you must remove all the binary fields from the field mapping.

The following table describes the Microsoft Fabric OneLake target properties that you can configure in a Target transformation:

Property	Description
Connection	Name of the target connection. Select a target connection or click **New Parameter** to define a new parameter for the target connection. If you want to overwrite the parameter at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties. When you switch between a non-parameterized and a parameterized Microsoft Fabric OneLake connection, the advanced property values are retained.
Target Type	Select Single Object or Parameter.
Object	Name of the target object. You can select an existing object or create a new target at runtime. When you select **Create New at Runtime**, enter a name for the target object and select the source fields that you want to use. By default, all source fields are used. The target name can contain alphanumeric characters. You can use only a period (.), an underscore (_), an at the rate sign (@), a dollar sign ($), and a percentage sign (%) special characters in the file name. Ensure that the headers or file data does not contain special characters. You can use parameters defined in a parameter file in the target name. When you select the Create Target option, you cannot parameterize the target at runtime. **Note:** When you write data to a flat file created at runtime, the target flat file contains a blank line at the end of the file.
Parameter	Select an existing parameter for the target object or click **New Parameter** to define a new parameter for the target object. The **Parameter** property appears only if you select Parameter as the target type. When you parameterize the target object, specify the complete object path including the file system in the default value of the parameter. If you want to overwrite the parameter at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties. Ensure that the parameter file is in the correct format.

Property	Description
Format	Specifies the file format that the Microsoft Fabric OneLake Connector uses to write data to Microsoft Fabric OneLake. Select one of the following file formats: - Avro - JSON - ORC - Flat - Parquet - Iceberg[1] - Delta. Write data to a delta lake. To use the delta file format in a mapping in advanced mode, see "Configuring Delta file format in a mapping in advanced mode" on page 22. Default is **None**. If you select **None** as the format type, Microsoft Fabric OneLake Connector writes data to Microsoft Fabric OneLake files in binary format. When you write binary data to a Microsoft Fabric OneLake target created at run time, ensure that the incoming field that contains the binary data is named **data.** If the field name differs, use field rules or an Expression transformation to change the field name to **data**. For more information, see "File formatting options" on page 20.
Operation	The target operation. You can insert data to a Microsoft Fabric OneLake target.
[1]Doesn't apply to mappings in advanced mode.	

Note: When you use the **Create Target** option and specify an object name with extension that does not match the `Format Type` under **Formatting Options**, the Secure Agent ignores the format type you specified under **Formatting Options**.

The following table describes the advanced target properties for Microsoft Fabric OneLake:

Advanced Target Property	Description
Workspace Override	Overrides the workspace specified in the connection.
Lakehouse Override	Overrides the lakehouse specified in the connection.
File Name Override	Overrides the source object at run time.
Write Strategy	Determines how you write to the target object. Select one of the following write strategy: - Overwrite. In a mapping and mapping in advanced mode, you can overwrite flat files, complex files, and delta files. - Append. In a mapping, you can append data to flat files and delta files. In a mapping in advanced mode, you can append data to flat files, complex files, and delta files. When you create a new target at runtime and use the append strategy, the mapping creates a new target file and writes the data to the file. The mapping appends data in subsequent runs. When you append data for mappings in advanced mode, the data is appended as a new part file in the existing target directory. The maximum size of data that you can append is 450 MB. Default is overwrite.

Advanced Target Property	Description
Compression Format	Compresses and writes data to the target based on the format you specify. Select one of the following options: - **None**. To write data to uncompressed files. - **Gzip**. To write data to flat files and Parquet files that use Gzip compression. - **Snappy**. To write data to Parquet files in Snappy compression format that uses the Snappy compression algorithm. When the task runs, the file extensions `.gz` or `.snappy` do not appear in target object name.
Forward Rejected Rows	Configure the transformation to either pass rejected rows to the next transformation or drop them.

Target time stamps

When you create a target at run time in a mapping, you can add time stamp information to the file name to show when the file is created.

When you specify the file name for the target file, include special characters based on Linux STRFTIME function formats that the mapping task uses to include time stamp information in the file name. The time stamp is based on the organization's time zone.

You cannot append time stamp information to the file name for mappings in advanced mode.

The following table describes some common STRFTIME function formats that you might use in a mapping or mapping task:

Special Character	Description
%d	Day as a two-decimal number, with a range of 01-31.
%m	Month as a two-decimal number, with a range of 01-12.
%y	Year as a two-decimal number without the century, with range of 00-99.
%Y	Year including the century, for example 2015.
%T	Applicable only to flat files. Time in 24-hour notation, equivalent to %H:%:M:%S.
%H	Hour in 24-hour clock notation, with a range of 00-24.
%I	Hour in 12-hour clock notation, with a range of 01-12.
%M	Minute as a decimal, with a range of 00-59.
%S	Second as a decimal, with a range of 00-60.
%p	Either AM or PM.

Note: For complex files, instead of %T you can use the equivalent %H_%M_%S.

Target partitioning

You can configure partitioning to optimize the mapping performance at run time when you write data to Microsoft Fabric OneLake. You can configure target partitioning only in mappings.

The partition type controls how the agent distributes data among partitions at partition points. With partitioning, the Secure Agent distributes rows of target data based on the number of threads that you define as partition.

For example, if there are three partitions in the source, the Secure Agent writes separate files for each partition in the Microsoft Fabric OneLake target in the following format:

```
<target>
<target_1>
<target_2>
```

Consider the following rules and guidelines for target partitioning:

- When you read from a directory with multiple partitions and configure target partitioning, the partition files are written to the target based on the number of partitions in the source. However, if you change the partitions in the source and run the mapping task again, ensure that you verify the existing partition files to avoid inconsistent data in the target.
- When you read data with multiple partitions and configure target partitioning, ensure that the target file name is unique and does not match the _part file name from any of the previous mapping runs. Otherwise, the target file might contain inconsistent data.
- You can use the append write strategy only for flat files.
- When you read from a parquet file and write to partitions in a Microsoft Fabric OneLake target, the recommended heap size for each partition is 0.5 GB.

File formatting options

Select the format of the Microsoft Fabric OneLake file and configure the formatting options.

The following table describes the formatting options for Avro, Parquet, JSON, ORC, Document, and delimited flat files:

Property	Description
Schema Source	The schema of the source or target file. Select one of the following options to specify a schema: - Read from data file. Imports the schema from a file in Microsoft Fabric OneLake. - Import from schema file. Imports the schema from a schema definition file in the agent machine.
Schema File	The schema definition file in the agent machine from where you want to upload the schema. You cannot upload a schema file when you create a target at runtime.

The following table describes the formatting options for flat files:

Property	Description
Flat File Type	The type of flat file. Select one of the following options: - Delimited. Reads a flat file that contains column delimiters. - Fixed Width. Reads a flat file with fields that have a fixed length. You must select the file format in the **Fixed Width File Format** option. If you do not have a fixed-width file format, click **New** > **Components** > **Fixed Width File Format** to create one.
Delimiter	Character used to separate columns of data in a delimited flat file. You can set values as comma, tab, colon, semicolon, or others. You cannot set a tab as a delimiter directly in the **Delimiter** field. To set a tab as a delimiter, you must type the tab character in any text editor. Then, copy and paste the tab character in the **Delimiter** field.
EscapeChar	Character immediately preceding a column delimiter character embedded in an unquoted string, or immediately preceding the quote character in a quoted string data in a delimited flat file. When you write data to Microsoft Fabric OneLake and specify a qualifier, by default, the qualifier is considered as the escape character. Else, the character specified as the escape character is considered.
Qualifier	Quote character that defines the boundaries of data in a delimited flat file. You can set qualifier as single quote or double quote.
Qualifier Mode	Specify the qualifier behavior when you write data to a delimited flat file. You can select one of the following options: - Minimal. Default mode. Applies qualifier to data enclosed within a delimiter value or a special character. - All. Applies qualifier to all data. - Non_Numeric. Not applicable. - All_Non_Null. Not applicable.
Disable escape character when a qualifier is set	Applicable to a Microsoft Fabric OneLake target. Select to disable the escape character when a qualifier is set. When you disable the escape character, the special characters not escaped and are considered as part of the data written to the target.
Code Page	Select the code page that the Secure Agent must use to read or write data to a delimited flat file. Select UTF-8 for mappings. Select one of the following options for mappings in advanced mode: - UTF-8 - MS Windows Latin 1 - Shift-JIS - ISO 8859-15 Latin 9 (Western European) - ISO 8859-3 Southeast European - ISO 8859-5 Cyrillic - ISO 8859-9 Latin 5 (Turkish) - IBM EBCDIC International Latin-1
Header Line Number	Specify the line number that you want to use as the header when you read data from a delimited flat file. Specify the value as 0 or 1. To read data from a file with no header, specify the value as 0.

Property	Description
First Data Row	Specify the line number from where you want the Secure Agent to read data in a delimited flat file. You must enter a value that is greater or equal to one. To read data from the header, the value of the **Header Line Number** and the **First Data Row** fields should be the same. Default is 1.
Target Header	Select whether you want to write data to a target that contains a header or without a header in the delimited flat file. You can select **With Header** or **Without Header** options. This property is not applicable when you read data from aMicrosoft Fabric OneLake source.
Distribution Column	Not applicable.
Max Rows To Preview	Not applicable.
Row Delimiter	Character used to separate rows of data. You can set values as `\r`, `\n`, and `\r\n`. This property is not applicable when you read data from a Microsoft Fabric OneLake source.

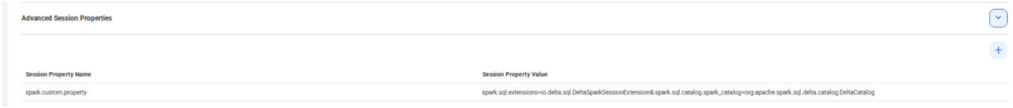
Configuring Delta file format in a mapping in advanced mode

You can use the Delta file format in mappings and mappings in advanced mode.

Before you use the Delta file format in mappings in advanced mode, perform the following steps:

1. In the mapping task, navigate to the **Advanced Session Properties** section on the **Runtime Options** tab.
2. Under **Advanced Session Properties**, click  and select **spark.custom.property** in the **Session Property Name** field.
3. In the **Session Property Value** field, set the following values:
   - spark.sql.extensions=io.delta.sql.DeltaSparkSessionExtension
   - spark.sql.catalog.spark_catalog=org.apache.spark.sql.delta.catalog.DeltaCatalog

   Separate the values with **&:** in the **Session Property Value** field.



Rules and guidelines for Delta file format

Consider the following rules and guidelines when you use the Delta file format to read from and write to a Delta Lake:

General guidelines

- You can read from and write to both `<Lakehouse_Name>/Files/` and `<Lakehouse_Name>/Tables/`.
- You can read data only when you select File as the **Source Type** in the advanced source properties.

Mappings

- Ensure that the source directory is a delta lake where data is stored in Parquet format and its corresponding transaction logs are stored in JSON format in a `_delta_log` folder. You can either select a JSON snapshot file or a Parquet file as a source object. If you select a Parquet file, the mapping processes the latest snapshot of the delta lake. If you select a JSON snapshot file, the mapping processes the corresponding state of the delta lake.
- The initial JSON file `00000000000000000000.json` contains the schema of the delta lake while the subsequent json files contain the transactional details.
- You can infer schema from the selected object only if the file is in JSON or text format.
-
- When you create a target at runtime, the specified target object name is not considered. Instead, the Delta Lake name is derived from the directory in the target Lakehouse path . For example, in the path `<Lakehouse_Name>\Files\Dir1\Dir2\test.parquet`, the Delta Lake name is taken from the directory `Dir2`.

Mappings in advanced mode

- You can only override the filename if the imported source object is a JSON file.
- When you perform an overwrite or append write strategy, the mapping creates a new data file and the transaction logs capture the transactions of the write strategy.

Rules and guidelines for Iceberg file format

Consider the following rules and guidelines when you use the Iceberg file format to read from or write to Apache Iceberg files:

- You can read data only when you select File as the **Source Type** in the advanced source properties.
- You can select the following as source objects:
  - JSON snapshot file in the metadata folder for time travel capabilities.
  - Any Parquet data file in the data folder to read the latest snapshot of the table.
  - Snapshot Avro file in the metadata folder to read the latest snapshot of the table.
    For example, `snap-1486404200109107576-1-5078c39c-8640-49ca-9985-da3359ddfca1.avro`
- You can infer schema from the selected object only if the file is in JSON or text format.
- When you create a target at runtime, the specified target object name is not considered. Instead, the Iceberg table name is derived from the directory that contains both the data and metadata of the table in the target Lakehouse path . For example, in the path `<Lakehouse_Name>\Files\Dir1\Dir2\test.parquet`, the Iceberg table name is taken from the directory `Dir2`. This `Dir2` directory contains both data and metadata folders.
- When you write to an existing target, you can only select Parquet files in the data folder.

Parameterization

You can parameterize the connection, objects, and the advanced runtime properties in mappings.

To parameterize the connection, objects, and the advanced runtime properties using a parameter file, create the parameters in the Parameters panel when you create a mapping. Then, define the parameters in the parameter file, place the parameter file in the following location, and run the mapping task:

<Informatica Cloud Secure Agent\apps\Data_Integration_Server\data\userparameters>

Consider the following rules and guidelines when you use parameterization:

General guidelines

- When you create a mapping with a parameterized target that you want to create at runtime, set the target field mapping to automatic.
- You cannot parameterize the field mapping.

Mappings

- When you use input parameters, specify the parameter name in the following format:
  - Format in a mapping task: $name$
  - Format in a parameter file: $name
- When you use in-out parameters, specify the parameter name in the following format in a mapping task or a parameter file: $$name.
- You cannot parameterize a Microsoft Fabric OneLake target created at runtime. Instead, you can specify the parameter in the Lakehouse Override to parameterize the target using a parameter file. Specify the parameter in the following format: $$name or $name.

Configure a lakehouse override in mappings

When you read from or write to Microsoft Fabric OneLake, you can specify the lakehouse override path in the advanced source and target properties to override the object path at runtime. The lakehouse override path can be an absolute or a relative path.

Consider the following guidelines when you override the object path:

Absolute Path

When you specify an absolute path in the lakehouse override property, the lakehouse override path overrides the object path.

Note: You must prefix the lakehouse name to the absolute path.

The following table exemplifies if you include the lakehouse name in the absolute path, the object path is correctly generated in the logs:

Object path	Lakehouse override path	Object path generated in logs
<<WorkSpace Name>>/ <<Lakehouse Name>>/ <<Files>>/<<Folder1>>/ Object.txt	<<Lakehouse Name>>/<<Files>>/ <<Folder2>>	<<<Lakehouse Name.lakehouse>>/ <<Files>>/<<Folder2>>

The following table exemplifies if you do not include the lakehouse name in the absolute path, the object path is incorrectly generated and the mapping fails:

Object path	Lakehouse override path	Object path generated in logs
<<WorkSpace Name>>/ <<Lakehouse Name>>/ <<Files>>/<<Folder1>>/ Object.txt	<<Folder2>>/<<Folder3>>	<<Folder2>>>.lakehouse/ <<Folder3>>

Relative Path

When you specify a relative path in the lakehouse override property, the lakehouse override path appends to the object path.

Note: You must not prefix the lakehouse name to the relative path.

The following table exemplifies if you do include the lakehouse name in the relative path, the object path is correctly generated in the logs:

Object path	Lakehouse override path	Object path generated in logs
<<WorkSpace Name>>/ <<Lakehouse Name>>/ <<Files>>/<<Folder1>>/ Object.txt	/<<Folder2>>/<<Folder3>>	<<Lakehouse Name.lakehouse>>/ <<Files>>/<<Folder1>>/ <<Folder2>>/<<Folder3>>

The following table exemplifies if you include the lakehouse name in the relative path, the object path is incorrectly generated and the mapping fails:

Object path	Lakehouse override path	Object path generated in logs
<<WorkSpace Name>>/ <<Lakehouse Name>>/ <<Files>>/<<Folder1>>/ Object.txt	/<<Lakehouse Name>>/<<Files>>/ <<Folder2>>	<<Lakehouse Name.lakehouse>>/ <<Files>>/<<Folder1>>/ <<Lakehouse Name>>/ <<Files>>/<<Folder2>>

Rules and guidelines for mappings

Consider the following rules and guidelines for mappings:

- To read and write complex files and to avoid java heap space error, set the JVM options for type DTM to increase the -Xms and -Xmx values in the system configuration details of the Secure Agent. The recommended -Xms and -Xmx values are 512 MB and 1024 MB respectively.
- When the source data contains special characters, the Secure Agent replaces the special characters with underscore (_) character in the target file.
- When you read zero-byte files or empty files, the Microsoft Fabric OneLake mapping fails with the following error:
  `error [okhttp3.internal.http2.StreamResetException: stream was reset: CANCEL]`

- When you use the Snappy compression format to write data to Microsoft Fabric OneLake, the mapping retains a **snappy-1.1.8****-libsnappyjava.so** file in the */tmp* directory on the agent machine after it runs successfully.
- When you run a mapping that reads from and writes to Microsoft Fabric OneLake and the source file name contains spaces along with special characters such as & and #, the data preview and mapping fails.
- When you read from a file or directory in Microsoft Fabric OneLake, ensure that the file name or directory name does not contain any emoji character.
- If the file name or lakehouse path has percentage (%), the data preview and mapping fails at runtime.
- To use the authenticated proxy server, add **dns_v4_first on** entry in `/etc/squid/squid.conf` file in the proxy server and restart the proxy server.
- If you configure the non-proxy host with HTTPS to access complex files and add dfs endpoints in the non-proxy host, you cannot bypass the proxy server. The endpoint addresses remain listed in the proxy logs.
- When you use the JVM options to configure the HTTPS proxy server and read and write complex files, the mapping fails at runtime.
- When you create a target at runtime, consider the following rules for the object name:
  - Always prefix the object name with the workspace name. For example, `<Workspace_Name>/<Lakehouse_Path>/Files/source.csv`.
  - If you specify a forward slash (/) in the Lakehouse Path connection property, you must specify the object name as `<Workspace_Name>/<Lakehouse_Path>/Files/source.csv`
  - If you specify the complete path in the Lakehouse Path connection property, you must specify the object name as `<Workspace_Name>/Files/source.csv`
- When you create a new target and select **Handle Special Characters** to append time stamp to the file name, the file name override is not honored in a mapping.
- When you parameterize a target object using a parameter file, ensure that the lakehouse name has a lakehouse extension. Else, the mapping fails.
- When you configure partial parameterization of lakehouse override in a mapping in advanced mode, the mapping fails.
- When you parameterize the connection and object and select the **Allow parameter to be overridden at runtime** option, you can only use a parameter file to override the default object and connection.
- When you partially parameterize the File Name Override value, specify the format of the parameter name as `$$<parameter_name>` in a mapping and `${$parameter_name}` in a mapping in advanced mode.
- When you write data to complex files where some of the fields are unmapped, the unmapped fields are written to the target with null data. However, when you write data to flat files, only the mapped fields are written to the target.
- When you run a mapping in advanced mode to read from Microsoft Fabric OneLake and write to Microsoft Azure Data Lake Storage Gen2, or read from Microsoft Azure Data Lake Storage Gen2 and write to Microsoft Fabric OneLake, ensure to configure different authentication types to access the source and target endpoints. For example, use Service Principal authentication to access Microsoft Fabric OneLake and Shared Key authentication to access the Microsoft Azure Data Lake Storage Gen2 account.
- When you create a mapping in advanced mode, you cannot preview data for the individual transformations. However, you can preview the data object when you create a mapping in advanced mode.
- When you run a parameterized mapping where you specify a relative lakehouse path in the target connection and root in the lakehouse path in the parameter file, the file path is incorrectly formed at runtime. To overcome this, replace the `<lakehouse_name/Files>` in the lakehouse path with the workspace name in the parameterized target object. For example, if the lakehouse path in the target

object path is `Lakehouse/Files/csv/dir/062_Onelake`, replace `Lakehouse/Files` with the workspace name in the parameter file.

- You cannot read zero-byte files when you run mappings in advanced mode.
- You can use the following special characters in a directory name or a file name:
  `` `, ~, !, @, #, $, ^, &, *, -, _, =, +, and () ``
- When you read objects from directories or subdirectories in Microsoft Fabric OneLake, ensure that the selected object exists in the source. If the object is deleted, the mapping fails with an object not found error.
- Consider the following guidelines when you use the document file format:
  - Ensure that the PDF files are not password-protected or compressed with Gzip compression.
  - Data Integration reads masked data in PDF files as-is and writes it to the target.
  - You can read up to 10 MB of text from a PDF file in a mapping.

CHAPTER 4

Data type reference

Data Integration uses the following data types in Microsoft Fabric OneLake mappings and mapping tasks:

- Microsoft Fabric OneLake native data types appear in the Source transformation and Target transformation when you choose to edit metadata for the fields.
- Transformation data types. Set of data types that appear in the transformations. These are internal data types based on ANSI SQL-92 generic data types, which the Secure Agent uses to move data across platforms. They appear in all transformations in a mapping.

When the Secure Agent reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When the Secure Agent writes to a target, it converts the transformation data types to the comparable native data types.

The following table lists the Microsoft Fabric OneLake data types that Data Integration supports and the corresponding transformation data types:

Microsoft Fabric OneLake Native Data Type	Transformation Data Type	Description
String	String	1 to 104,857,600 characters

Avro data types and transformation data types

Avro file data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the Avro file data types that the Secure Agent supports and the corresponding transformation data types:

Avro Data Type	Transformation Data Type	Range and Description
Boolean	Integer	1 or 0 True is equivalent to the integer 1 and False is equivalent to the integer 0.
Bytes	Binary	Precision 4000
Double	Double	Precision 15
Float	Double	Precision 15

Avro Data Type	Transformation Data Type	Range and Description
Int	Integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
Long	Bigint	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 Precision 19, scale 0
Null	Integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
String	String	1 to 104,857,600 characters Precision 4000

JSON file data types and transformation data types

JSON file data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the JSON file data types that the Secure Agent supports and the corresponding transformation data types:

JSON Data Type	Transformation Data Type	Range and Description
boolean	integer	The default transformation type for boolean is integer. You can specify string data type with values of True and False. True is equivalent to the integer 1 and False is equivalent to the integer 0.
Number (double)	double	-1.79769313486231570E+308 to +1.79769313486231570E+308. Precision 15.
Number (float)	double	-1.79769313486231570E+308 to +1.79769313486231570E+308. Precision 15.
Number (int)	integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
Number (long)	bigint	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 Precision 19, scale 0.
string	string	1 to 104,857,600 characters. Precision 4000

ORC data types and transformation data types

ORC file data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the ORC file data types that the Secure Agent supports and the corresponding transformation data types:

ORC File Data Type	Transformation Data Type	Range and Description
BigInt	BigInt	-9223372036854775808 to 9,223,372,036,854,775,807
Boolean	Integer	1 or 0 True is equivalent to the integer 1 and False is equivalent to the integer 0.
Char	String	1 to 104,857,600 characters
Date	Date/Time	Jan 1, 1753 A.D. to Dec 31, 4712 A.D. (precision to microsecond)
Double	Double	Precision of 15 digits
Float	Double	Precision of 15 digits
Integer	Integer	-2,147,483,648 to 2,147,483,647
SmallInt	Integer	-32,768 to 32,767
String	String	1 to 104,857,600 characters Precision 4000
Timestamp	Date/Time	1 to 19 characters Precision 19 to 26, scale 0 to 6
TinyInt	Integer	-128 to 127
Varchar	String	1 to 104,857,600 characters

Flat file data types and transformation data types

Flat file data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the flat file data types that the Secure Agent supports and the corresponding transformation data types:

Flat file data type	Transformation data type for mappings	Transformation data type for mappings in advanced mode	Range and description
Nstring*	Text	Text	1 to 104,857,600 characters
Number*	Decimal	Double	Precision from 1 through 28 digits, scale from 0 through 28 digits
String*	String	String	1 to 104,857,600 characters. Precision 256.
*You must select the **Schema Source** as **Import from schema file** to read data of Number, String, or Nstring data type.			

Parquet data types and transformation data types

Parquet file data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the Parquet file data types that the Secure Agent supports and the corresponding transformation data types:

Parquet data type	Transformation data type	Range and description
Boolean	Integer	1 or 0 True is equivalent to the integer 1 and False is equivalent to the integer 0.
Byte_Array	Binary	Arbitrarily long byte array
Date	Date/Time	January 1, 0001 to December 31, 9999.
Decimal	Decimal	Precision 1 to 28 digits, scale 0 to 28. You cannot use decimal values with precision greater than 28.
Double	Double	Precision 15
Float	Double	Precision 15
Int32	Integer	-2,147,483,648 to +2,147,483,647
Int64	Bigint	-9,223,372,036,854,775,808 to +9,223,372,036,854,775,807 8-byte signed integer

Parquet data type	Transformation data type	Range and description
Int96	Binary	12-byte signed integer When you configure a mapping in advanced mode, the data preview displays the time zone of the Secure Agent machine.
String	String	1 to 104,857,600 characters Precision 4000
Time	Date/Time	Time of the day. Precision to microsecond.
Timestamp	Date/Time	January 1, 0001 00:00:00 to December 31, 9999 23:59:59.997. Precision to microsecond. You cannot set the precision to nanoseconds.

The Parquet schema that you specify for the Parquet file must be in smaller case. Parquet does not support case-sensitive schema.

Parquet timestamp data type support

You can use the following Timestamp data types for Parquet file format:

Timestamp Data type	Mappings	Mappings in advanced mode
Timestamp_micros	Yes	No
Timestamp_millis	Yes	No
Time_millis	Yes	No
Time_micros	Yes	No
int96	Yes	Yes

You cannot use the following Timestamp data types for Parquet file format:

- Timestamp_nanos
- Time_nanos
- Timestamp_tz

Delta data types and transformation data types

Delta data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the Delta data types that the Secure Agent supports and the corresponding transformation data types:

Delta data type	Transformation data type	Range and description
Boolean	Boolean	Represents Boolean values.
Binary	Binary	Represents byte sequence values.
Byte	Integer	-128 to 127
Date	Date/Time	January 1, 0001 to December 31, 9999.
Decimal	Decimal	Precision 1 to 24 digits, scale 0 to 24. You cannot use decimal values with precision greater than 24.
Double	Double	Represents 8-byte double-precision floating point numbers.
Float	Double	Represents 4-byte single-precision floating point numbers
Integer	Integer	-2147483648 to 2147483647
Long	Bigint	-9223372036854775808 to 9223372036854775807
Short	Integer	-32768 to 32767
String	String	1 to 104,857,600 characters Precision 4000.
Timestamp	Date/Time	January 1, 0001 00:00:00 to December 31, 9999 23:59:59.997, with precision up to microseconds. You cannot set the precision to nanoseconds.

Iceberg data types and transformation data types

Iceberg data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the Iceberg data types that the Secure Agent supports and the corresponding transformation data types:

Iceberg data type	Transformation data type	Range and description
Binary	Binary	Represents byte sequence values.
Boolean	Int	1 or 0
Date	Date/Time	October 15, 1582 to December 31, 9999.
Decimal	Decimal	Precision 1 to 28 digits, scale 0 to 28.
Double	Double	Precision 15.
Float	Double	Precision 7.
Fixed(L)	Binary	Fixed length value.
Long	Bigint	-9,223,372,036,854,775,808 to +9,223,372,036,854,775,807
Int	Int	-2,147,483,648 to +2,147,483,647
String	String	4000
Time[1]	Bigint	Time of the day.
Timestamp [1]	Date/Time	October 15, 1582 00:00:00 to December 31, 9999 23:59:59.997. Precision to microsecond.
Timestamptz	Date/Time	Date and time with timezone.
UUID[1]	Binary	
UUID[2]	String	16
[1] Doesn't apply to mappings in advanced mode.		
[2] Applies only to mappings in advanced mode.		

CHAPTER 5

Troubleshooting

Use the following section to troubleshoot errors in mappings.

Troubleshooting a mapping or mapping task

Mapping failed with a Java heap space error

When you read from or write to large data sets in Microsoft Fabric OneLake, certain mappings might fail with the following error:

```
[ERROR] java.lang.OutOfMemoryError: Java heap space
```

You must increase the heap size to run the mappings successfully. The recommended heap size is 1 GB.

Perform the following steps to configure the JVM options in the Secure Agent to increase the memory for the Java heap size:

1. Select **Administrator** > **Runtime Environments**.
2. On the **Runtime Environments** page, select the Secure Agent for which you want to increase memory from the list of available Secure Agents.
3. In the upper-right corner, click **Edit**.
4. In the **System Configuration Details** section, select **Data Integration Server** as the service and **DTM** as the type.
5. Edit the **JVMOption** field and set the value to **-Xmx1024m**.

   Note: The recommended heap size is 1 GB. You can increase the heap size based on the data you want to process.
6. Click **Save**.

Troubleshooting a mapping in advanced mode

Mapping configured to read or write Date and Int96 data types for Parquet file fails

A mapping configured to read from or write to a Parquet file fails in the following cases:

- Data is of the Date data type and the date is less than 1582-10-15.
- Data is of the Int96 data type and the timestamp is less than 1900-01-01T00:00:00Z.

To resolve this issue, specify the following spark session properties in the mapping task or in the custom properties file for the Secure Agent:

- `spark.sql.parquet.int96RebaseModeInWrite=LEGACY`
- `spark.sql.parquet.datetimeRebaseModeInWrite=LEGACY`
- `spark.sql.parquet.int96RebaseModeInRead=LEGACY`
- `spark.sql.parquet.datetimeRebaseModeInRead=LEGACY`

Mapping fails when you write data to flat files

When you write data to flat files in Microsoft Fabric OneLake, the mapping fails with the following error: `Caused by: java.io.IOException: Failed to rename VersionedFileStatus{VersionedFileStatus`

To resolve this issue, perform the following steps:

1. Under **Spark Session Properties**, click **Add** and select `spark.custom.property` in the **Session Property Name** field.
2. In the **Session Property Value** field, specify the following property and set its value to `DIRECT`: `infaspark.flatfile.writer.committer.name`

   For example, `infaspark.flatfile.writer.committer.name=DIRECT`

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